

Qatar-2b

R-2004

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_140623 · created 2026-08-07T16:19:28+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456831.683557 +/- 5.9e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2282 +/- 0.0099
Transit depth (Rp/Rs)^2	5.21 +/- 0.45 [%]
Orbital Inclination (inc)	85.83 +/- 1.42
Airmass coefficient 1 (a1)	1.11 +/- 0.031
Airmass coefficient 2 (a2)	-0.062 +/- 0.02
Scatter in the residuals of the lightcurve fit is	2.79 %
Best Comparison Star	#1 - [373, 279]
Optimal Aperture	2.97
Optimal Annulus	14.29
Transit Duration (day)	0.0505 +/- 0.0022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.2282 $\pm$ 0.0099 vs 0.16327 (deviation 4.19 σ)
Duration fit vs published	0.0505 d vs 0.0767 d (deviation 7.61 σ)
Mid-time O-C	0.3 min
Depth SNR	14.7
Residual scatter	2.79 %

Light curve

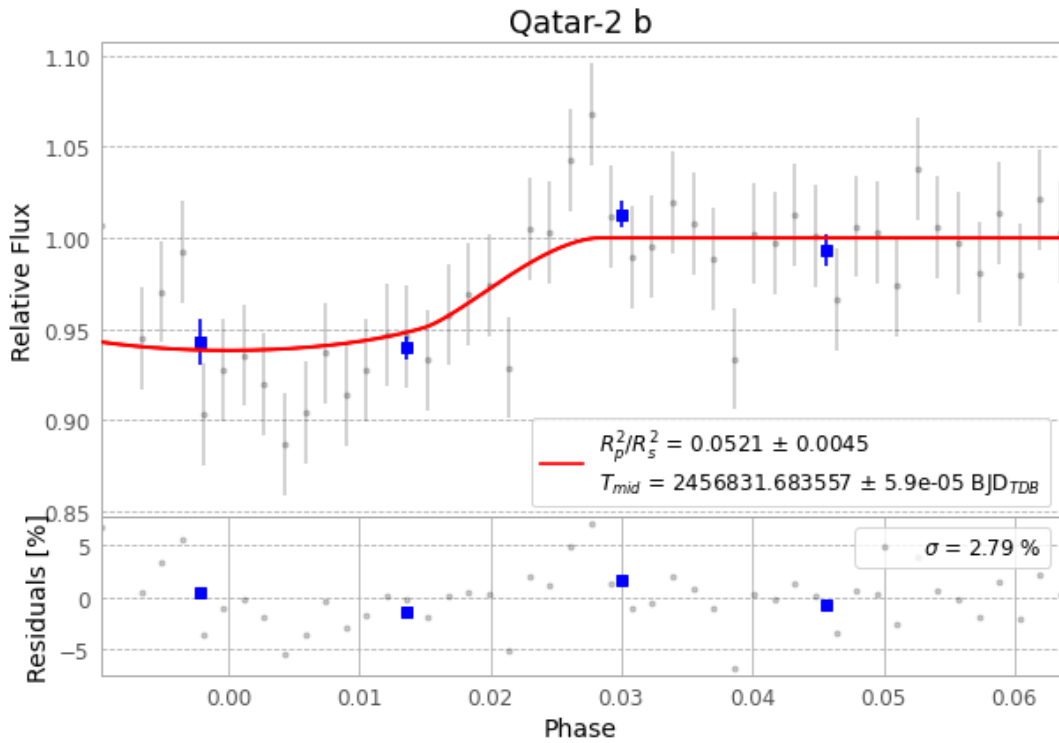


Figure 1 — FinalLightCurve_Qatar-2 b_2014-06-22.png (SHA-256 a43c86d4a123f24b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [388, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 35d14a5323fc5ad2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

48 FITS frames (32 MB), Qatar-2140623040012.FITS → Qatar-2140623062113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-140623.log (e33379b45e33...), FinalLightCurve_Qatar-2 b_2014-06-22.png (a43c86d4a123...), FinalLightCurve_Qatar-2 b_2014-06-22.csv (ff01b772b4f7...)

Compute resources

Wall time 97.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 16:19Z.